

PDF → HTML Reconstruction Benchmark

20-page visual fidelity test document

Purpose: benchmark whether the application can convert a PDF into an editable HTML representation that visually matches the source page. The document deliberately progresses from basic text to complex mixed layouts.

Evaluation dimensions

Dimension	What to inspect
Geometry	Page size, coordinates, margins, region positions
Typography	Font size, weight, line height, alignment, wrapping
Structure	Paragraphs, headings, lists, tables, captions, columns
Assets	Images remain original/correctly cropped and positioned
Editability	Text and tables are actual HTML elements, not screenshots
Visual fidelity	Rendered HTML should resemble the PDF page, not a redesign

Difficulty schedule

Pages 1–5: easy baseline. **6–9:** mixed content. **10–13:** hard structural layouts. **14–17:** very hard vector, nested, asymmetric and dense layouts. **18–20:** extreme composite pages.

EXPECTED APPLICATION BEHAVIOUR — PAGE 1

- Produce one editable HTML page corresponding to this PDF page, using a fixed page canvas with the same aspect ratio.
- Do not replace the page with a generic flowing article layout.
- Preserve all visible text, region boundaries, spacing and visual hierarchy.

1. Basic Typography

Text extraction baseline

This page tests the simplest possible case: title, headings, paragraphs, emphasis and multiple lines. The PDF-to-HTML system should use actual text nodes rather than rasterizing the page.

Bold: The important phrase is emphasized. *Italic:* this phrase is slanted. Underline: this phrase is underlined.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Integer vitae sem vel augue aliquet tempor. Suspendisse potenti. Curabitur malesuada, justo at fermentum luctus, nibh purus pretium magna, a efficitur lacus neque sed arcu.

Second paragraph with a deliberately different line length. The line breaks are created by the page width and typography, not by explicit HTML line-breaks.

LEFT	CENTER	RIGHT
Alpha	Beta	Gamma
123	456	789

EXPECTED APPLICATION BEHAVIOUR — PAGE 2

- Extract every text item accurately and preserve bold/italic/underline where present.
- Keep the three alignment modes distinct.
- The small table must remain a real editable HTML table with the same width and alignment.

2. Simple Table

Employee directory

A regular rectangular table is used as the baseline for table detection and reconstruction.

Employee ID	Name	Department	Status	Score
E-001	Aarav Mehta	Engineering	Active	94
E-002	Maya Rao	Design	Active	91
E-003	Daniel Kim	Research	On leave	88
E-004	Sara Thomas	Finance	Active	97
E-005	Noah Wilson	Operations	Active	86
E-006	Isha Patel	Engineering	Active	93
E-007	Liam Chen	Research	Active	95

Table note

The table intentionally contains different text lengths and numeric alignment. It should not be converted into a single image.

EXPECTED APPLICATION BEHAVIOUR — PAGE 3

- Detect the complete table boundary.
- Recreate rows, columns, borders, cell padding and numeric alignment.
- Do not OCR the table into a paragraph or flatten it into an image.

3. Lists and Indentation

Conversion checklist

1. Extract page dimensions and establish the coordinate system.
2. Identify headings and paragraph regions.
3. Detect nested list indentation.
4. Preserve line spacing and paragraph spacing.
5. Export the result as editable HTML.

Nested requirements

- Primary requirement: visual fidelity
 - Text remains selectable
 - Tables remain editable
 - Images remain independent assets
- Secondary requirement: clean semantic HTML

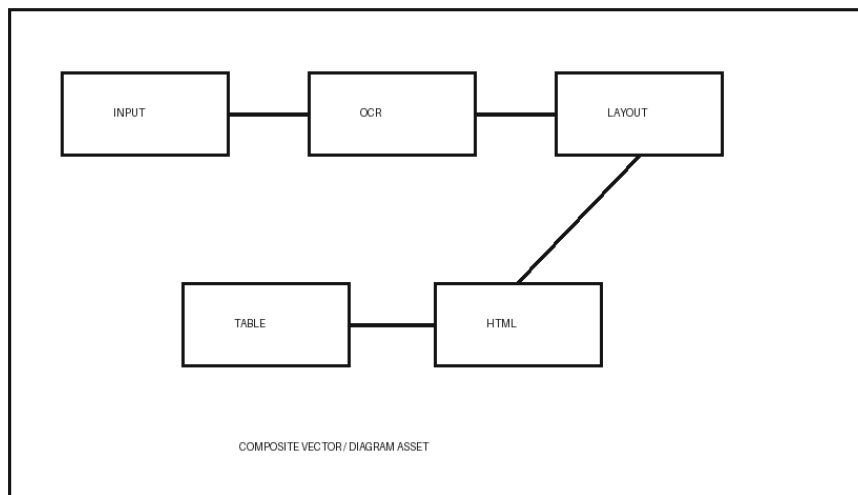


Figure 4. A small diagram is introduced while the page remains structurally simple.

EXPECTED APPLICATION BEHAVIOUR — PAGE 4

- Preserve list indentation and the distinction between parent and child items.
- Extract the diagram as a separate visual asset or reconstruct its vector geometry without moving it.
- Maintain the caption directly beneath the figure.

4. Quote, Callout and Image

Mixed but still easy

A quote, explanatory text and a raster figure coexist on this page.

CALLOUT

This block is intentionally visually distinct from normal body text.

“Accuracy is achieved by preserving what was actually there, not by redesigning it.”

ORIGINAL RASTER FIGURE

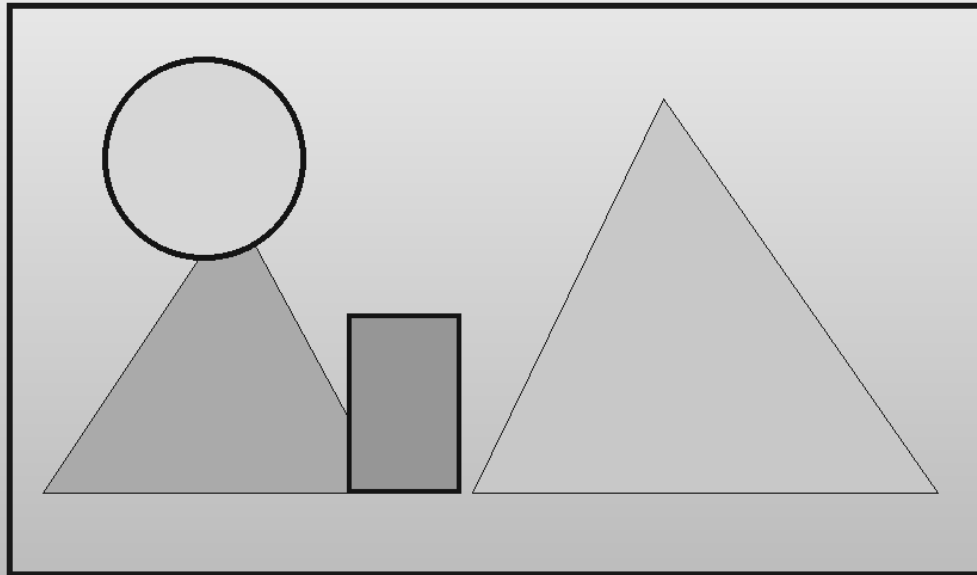


Figure 5. Original raster asset. The benchmark expects this image to appear at approximately the same size and position.

EXPECTED APPLICATION BEHAVIOUR — PAGE 5

- Keep the callout as an independent region.
- Keep the quotation centered and separate from the surrounding paragraph flow.
- Extract or reference the original raster image; do not generate a replacement image.
- Preserve the figure caption and its association with the image.

5. Image Placement and Caption

Raster asset test

This page tests whether the application can identify an image, extract the original pixels, and place the asset at the correct coordinates.

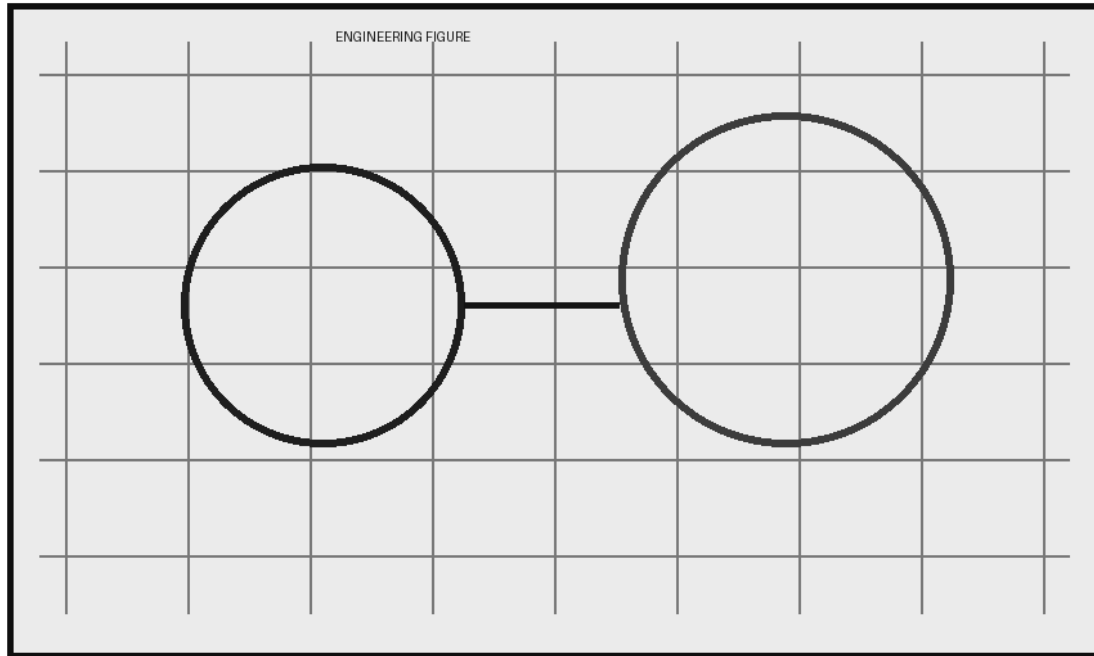


Figure 6. Engineering-style raster figure containing lines, circles and multiple spatial regions.

Property	Expected
Asset type	Raster image
Editing	Image remains an independent element
Geometry	Same aspect ratio
Placement	Same relative location

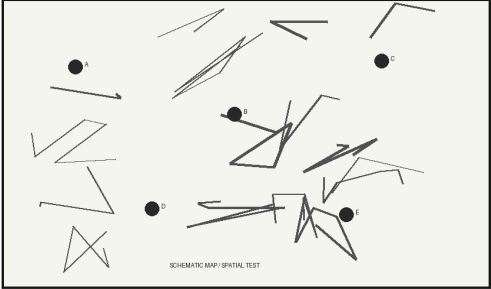
EXPECTED APPLICATION BEHAVIOUR — PAGE 6

- The image should be extracted without unnecessary recompression or distortion.
- The table below must remain editable and must not be merged with the image.
- Image and table should remain independently selectable in the HTML editor.

6. Two-Section Document

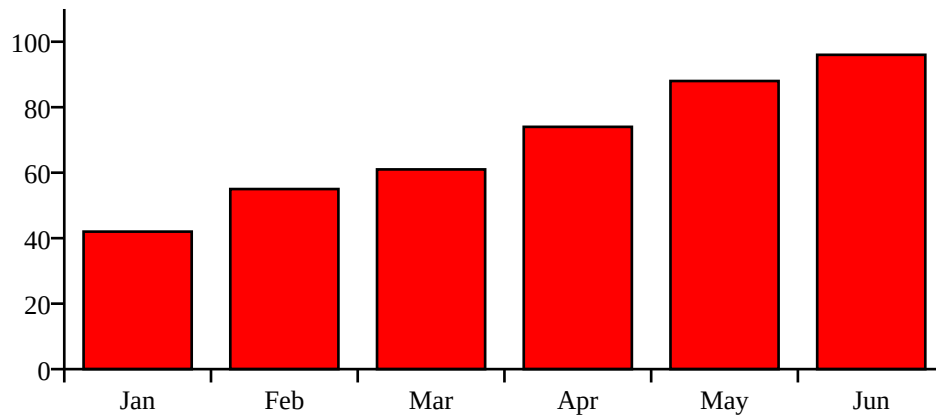
Overview and specification

The first section explains the benchmark. The second section contains a compact specification table. A small map-like figure is positioned beside the explanatory content.

OVERVIEW Document reconstruction requires geometry, text recognition and asset extraction.	SPATIAL FIGURE 
EXPECTED APPLICATION BEHAVIOUR — PAGE 7 • Preserve the two-column relationship inside the boxed region. • The map image is a separate asset and must not be mistaken for text. • Maintain the independent section hierarchy and table/figure relationships.	

7. Chart and Supporting Text

Monthly processing throughput



Interpretation. Throughput increases steadily from January through June. The chart is a visual object with its own geometry.

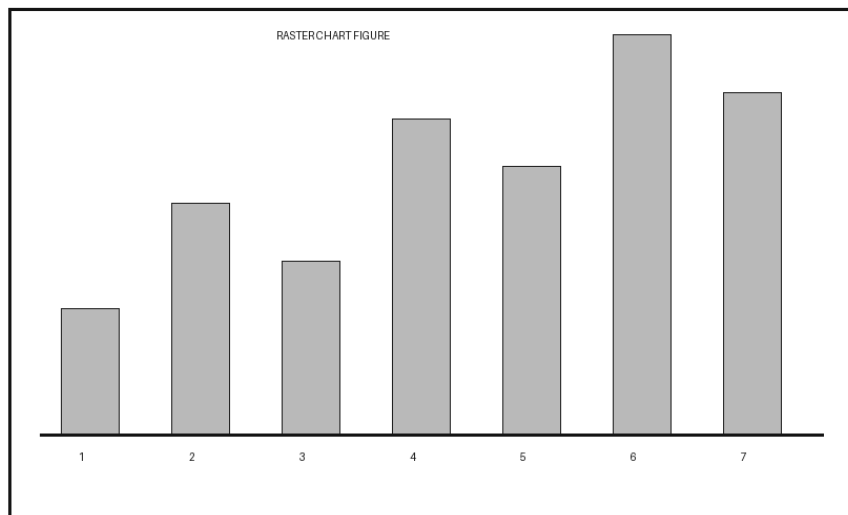


Figure 8. Additional raster chart used to test mixed chart/paragraph/image placement.

EXPECTED APPLICATION BEHAVIOUR — PAGE 8

- Preserve the vector chart as a distinct region if vector extraction is available; otherwise preserve its appearance as a chart asset.
- Do not allow the supporting paragraph to overlap the chart.
- The second chart image must remain independent and correctly positioned.

8. Three-Column Information Page

Independent regions

TEXT OCR should preserve line breaks, paragraph spacing and local alignment. A second paragraph belongs only to this column.	TABLE Tables should be represented by real editable cells. Column widths matter.	IMAGE Raster regions should come from the original PDF whenever possible. The image has its own bounding box.
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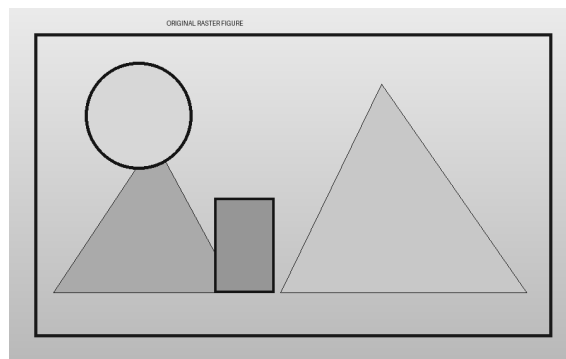


Figure 9. Image placed below the three-column region.

EXPECTED APPLICATION BEHAVIOUR — PAGE 9

- Keep the three top columns independent.
- Do not concatenate their text into a single reading stream when reconstructing the page visually.
- Keep the image below the columns and preserve its caption.

9. Multi-Column Research Layout

Difficulty begins: two columns + sidebar

Introduction

Document pages often use multiple columns. A reconstruction system must not merge text from adjacent columns merely because OCR returned a continuous token sequence.

Method

Spatial relationships, column gutters and region boundaries determine the layout.

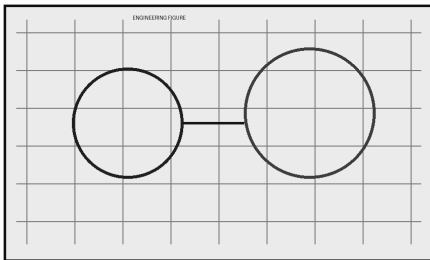


Figure 10A

Results

The right column contains independent material and must remain visually separated from the left article.

Observation

Column widths and vertical alignment are part of the page structure.

Metric	Value
Precision	0.98
Recall	0.96
F1	0.97

SIDEBAR

A narrow auxiliary region tests whether the system distinguishes side content from the main reading flow.



10B

EXPECTED APPLICATION BEHAVIOUR — PAGE 10

- Detect three distinct vertical regions: main column, second column, and narrow sidebar.
- Do not merge sidebar text into the main reading order visually.
- Preserve the image, mini-table, captions and column gutters.

10. Complex Table with Merged-Style Grouping

Quarterly resource allocation

Division	Q1	Q2	Q3	Q4	Annual
Engineering	120	135	148	160	563
Design	80	92	96	104	372
Research	95	105	118	127	445
Operations	60	72	74	83	289
Total	355	404	436	474	1669

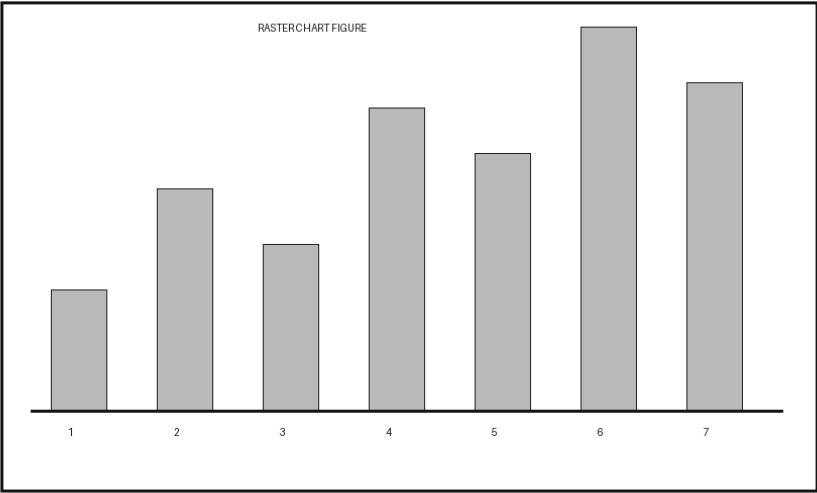


Figure 11. A chart is intentionally placed below a dense table.

- EXPECTED APPLICATION BEHAVIOUR — PAGE 11**
- Reconstruct the table as editable cells and preserve the emphasized total row.
 - Keep the chart as an independent object below the table.
 - Preserve numeric alignment, table width and the vertical relationship between table and chart.

11. Formula and Figure Composition

Technical page

Consider the following relationships:

$$E = mc^2$$
$$a_n = a_1 + (n - 1)d$$

The formulas are separate visual regions, not ordinary prose.

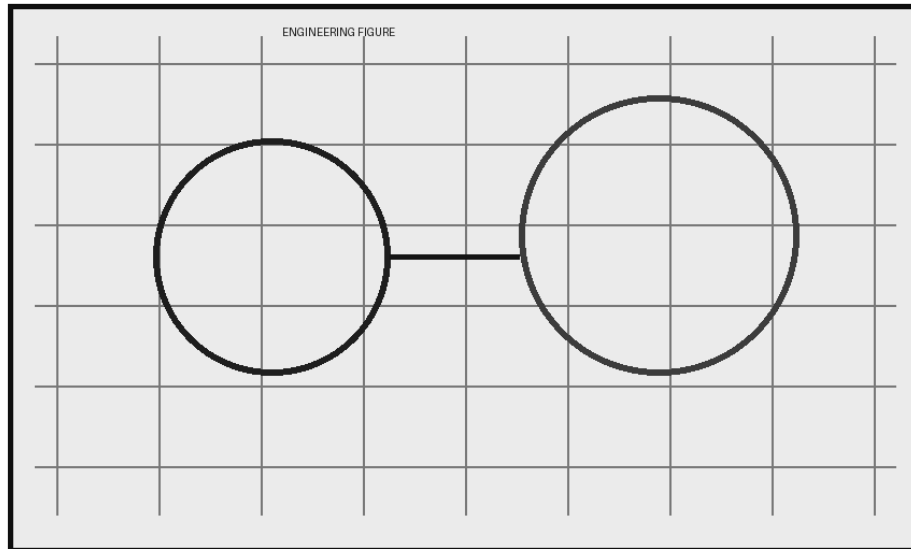


Figure 12. Technical raster figure positioned after formula regions.

Additional explanation: mathematical content may require specialized extraction or preservation as a vector/image region if exact editable mathematical markup is unavailable.

EXPECTED APPLICATION BEHAVIOUR — PAGE 12

- Keep each formula visually separate and correctly centered.
- Do not let OCR destroy superscripts/subscripts or merge formulas into surrounding prose.
- Preserve the technical figure as an independent asset.

12. Captioned Figure with Dense Text

Experimental setup

This page intentionally surrounds a figure with explanatory content. The figure, caption, and paragraphs form separate regions with explicit spatial relationships.

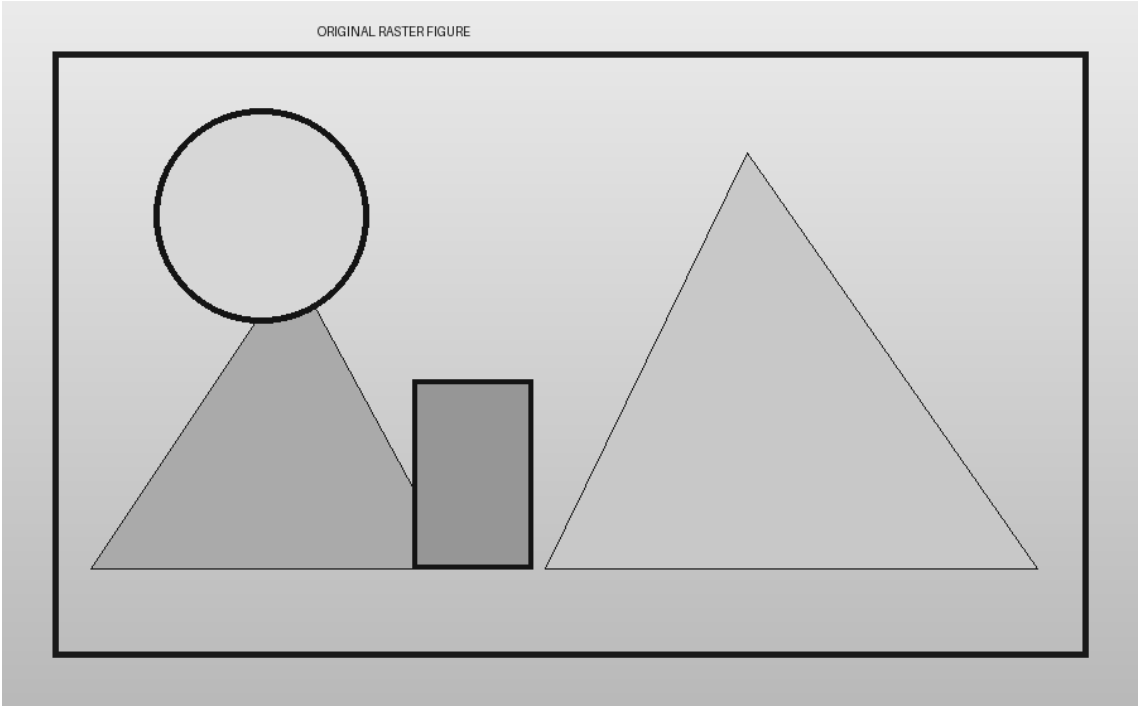


Figure 13. Benchmark raster figure showing geometric primitives and multiple internal shapes.

The paragraph below the figure is a new text region. Its vertical position should be determined by the figure and caption rather than by ordinary web flow.

Parameter	Value	Unit
Scale	0.60	relative
Resolution	300	dpi
Crop	Original	—

EXPECTED APPLICATION BEHAVIOUR — PAGE 13

- Keep the figure, caption, paragraph and table as four separate regions.
- Preserve the exact figure aspect ratio and approximate coordinates.
- The parameter table must remain editable.

13. Nested Information Structure

Table inside table

PROCESS	DETAIL		
Stage A	Extract page-level geometry and native assets.		
Stage B	Input	Processing	Output
	PDF	Extraction	Text
	Scan	OCR	Regions
Stage C	Render elements into a fixed coordinate canvas.		
Stage D	Compare the rendered HTML page against the original PDF.		

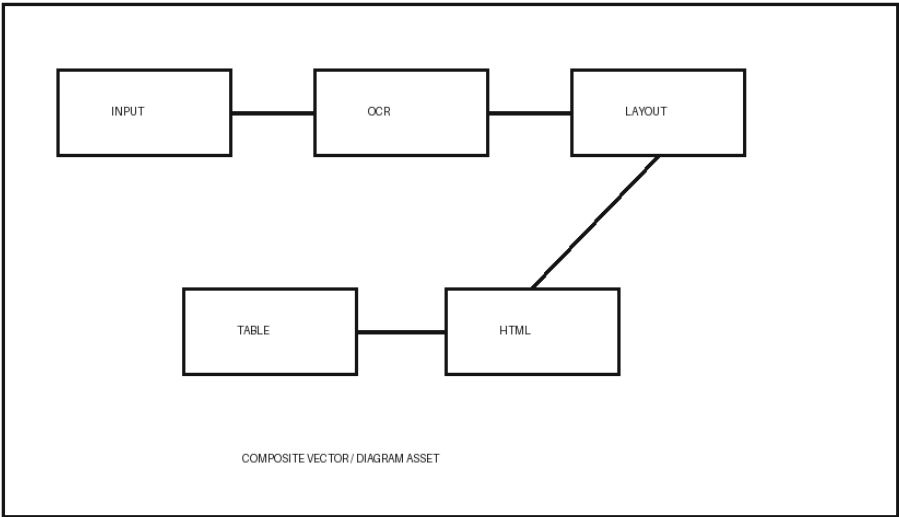


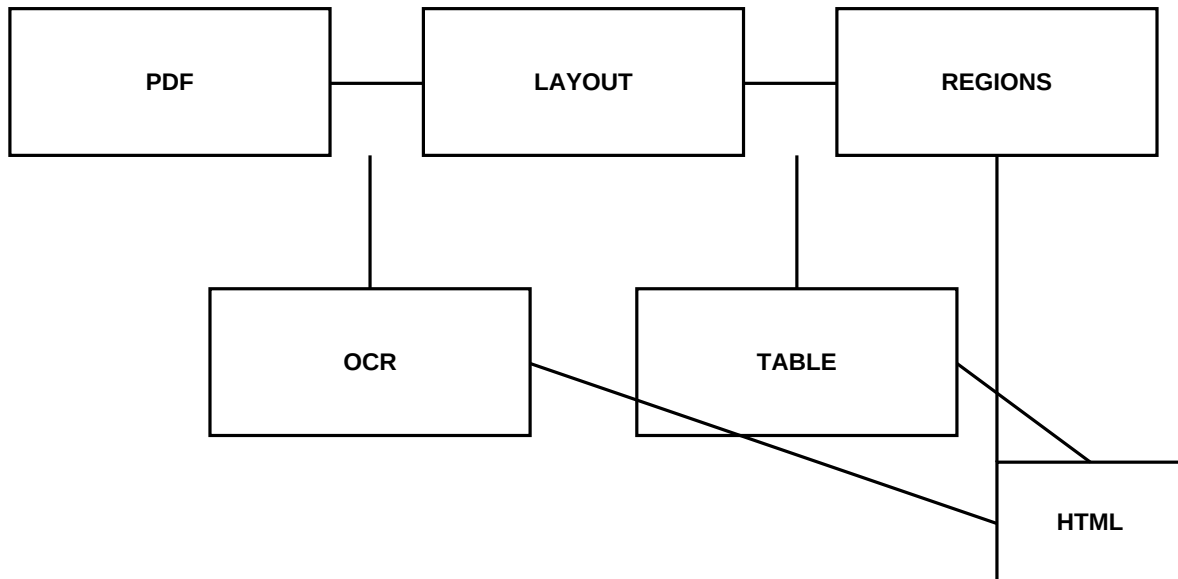
Figure 14. Diagram below a nested table.

EXPECTED APPLICATION BEHAVIOUR — PAGE 14

- Detect the outer table and inner table as nested structural objects.
- Do not flatten the inner table into text.
- Keep the diagram below the table and preserve all boundaries.

14. Vector Diagram

Coordinate-preserving processing flow



The diagram tests vector-like rectangles, lines, arrows, alignment and mixed directional relationships.

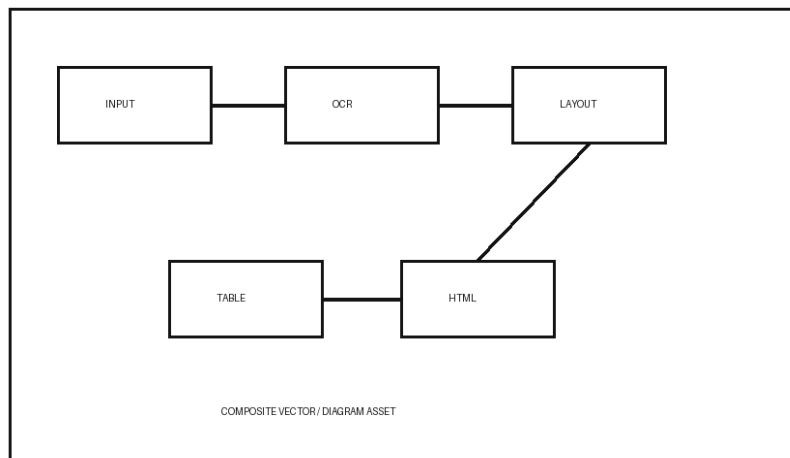


Figure 15. Raster copy of a similar diagram, creating an additional vector-vs-raster test.

EXPECTED APPLICATION BEHAVIOUR — PAGE 15

- Preserve the vector diagram's geometry rather than treating every shape as a paragraph.
- The raster copy is a separate asset and should not be confused with the vector diagram.
- Maintain exact relative placement between the two diagrams.

15. Dense Mixed Layout

Asymmetric information blocks

PRIMARY FINDING The largest region contains the main finding and should dominate the page.	METRIC 97.4%
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Observation A A narrow note beside a larger content block.	Observation B An independent region with its own spacing.	Observation C A third block that must not merge with its neighbors.
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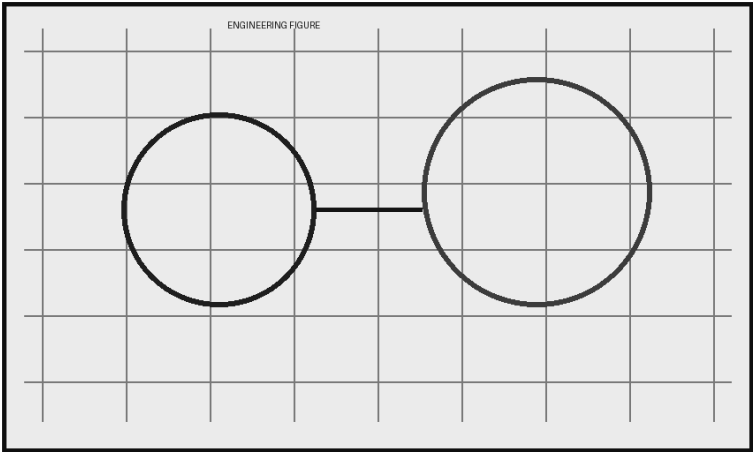


Figure 16. Dense technical asset under asymmetric content blocks.

- EXPECTED APPLICATION BEHAVIOUR — PAGE 16**
- Preserve the unequal widths and independent block boundaries.
 - Do not convert the page into a uniform grid.
 - The technical image must remain below the content blocks and retain its aspect ratio.

16. Asymmetric Editorial Layout

Article + narrow reference rail

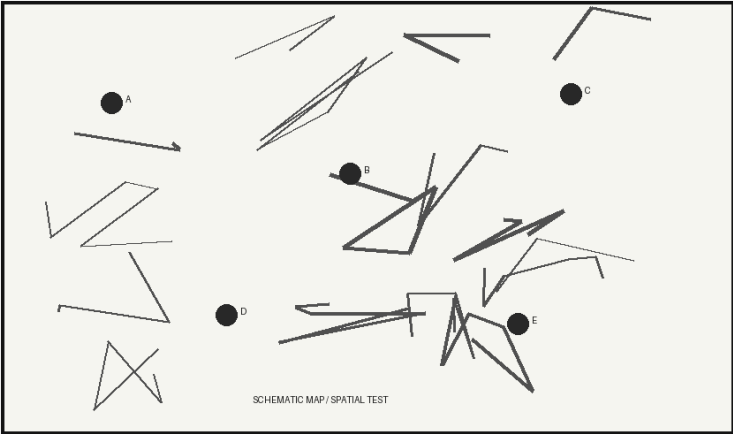
REFERENCE A narrow vertical region containing secondary notes, dates and labels. 2026 Benchmark v2	MAIN ARTICLE This is the dominant text region. It contains several lines of prose and should maintain a stable width while the narrow reference rail remains independent. A reconstruction that simply flows all content into one column will visibly fail this page. Continued Additional content appears below the main article and should begin at its original vertical coordinate.
FIGURE NOTE Small secondary annotation.	

Figure 17. Map/diagram asset inside the lower-right region.

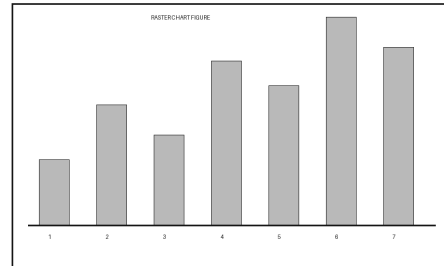
- EXPECTED APPLICATION BEHAVIOUR — PAGE 17**
- Preserve the narrow reference rail and the dominant article column.
 - Keep the lower-right figure separate from the article text.
 - Do not allow normal HTML reflow to change the intended asymmetric geometry.

17. Complex Dashboard-Like Page

Performance overview

Processed	Accuracy	Tables	Images
12,480	98.2%	1,126	3,842

CHART



NOTES

A compact operational summary sits beside the chart.

Class	F1
Text	0.99
Table	0.97
Image	1.00

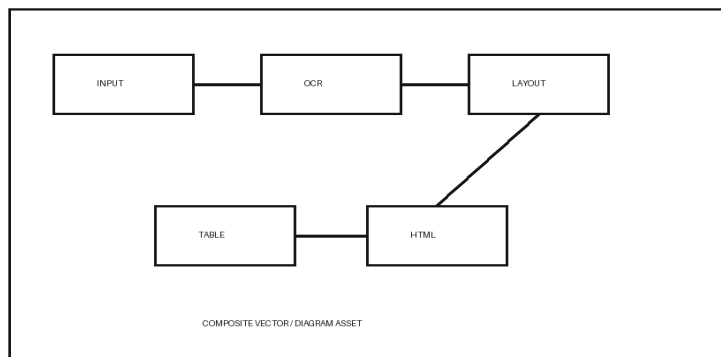


Figure 18. Diagram added to increase mixed-object density.

EXPECTED APPLICATION BEHAVIOUR — PAGE 18

- Keep all four KPI cards visually independent.
- Preserve the chart, notes, nested table and diagram as separate regions.
- Do not flatten the dashboard into one image or one flowing text column.

18. High-Complexity Research Page

Dense article + sidebar + figure + table

Abstract

This region contains dense prose. It should remain separate from the figure and right-hand note.

1. Background

Several paragraphs form a continuous article region. The renderer must preserve line widths and vertical rhythm.

2. Results

Results contain a compact table followed by explanatory text.

Class	F1	Recall
Text	0.99	0.98
Table	0.97	0.95
Image	1.00	0.99

KEY METRIC

0.982

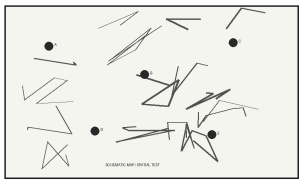


Figure 19A

NOTE

The right column is not part of the article reading flow.

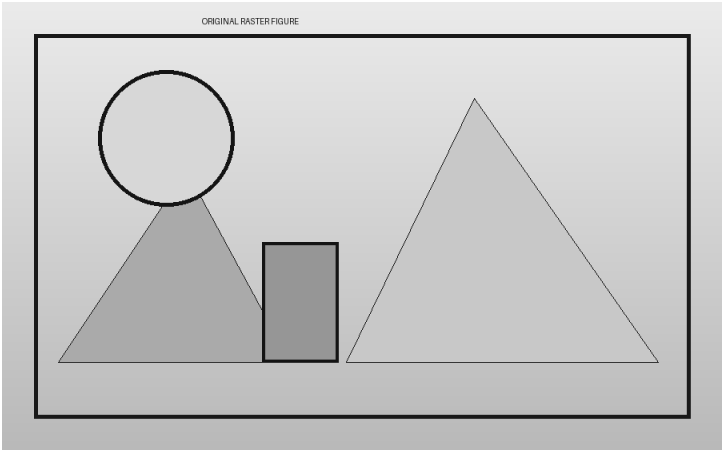


Figure 19B. Secondary raster figure beneath the multi-column content.

EXPECTED APPLICATION BEHAVIOUR — PAGE 19

- Keep the article column, sidebar, nested table, sidebar figure and bottom figure distinct.
- Preserve the reading order visually without collapsing spatial layout.
- This page should expose errors in column detection, asset placement and table reconstruction.

19. Final Extreme Reconstruction Test

Composite page — all major region types

TITLE REGION Prominent heading with subtitle and a separate status region.	STATUS PASS
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Region	Type	Difficulty	Editable
A	Native text	Medium	Yes
B	Table	Hard	
C	Raster image	Easy	
D	Vector diagram	Very hard	Partial

TEXT

TABLE

IMAGE

FORMULA

SVG

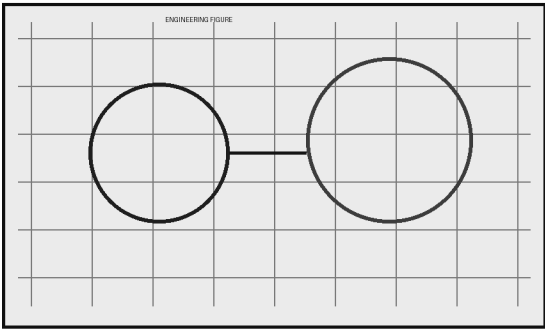


Figure 20A. Raster asset in the final composite page.

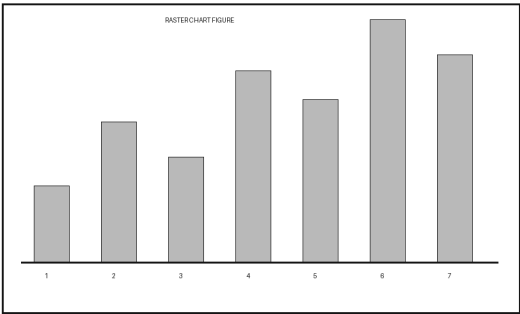


Figure 20B. Second raster asset. The two figures must remain independent.

EXPECTED APPLICATION BEHAVIOUR — PAGE 20

- Preserve every independent region: title, status, table, vector diagram, two raster figures and surrounding text.
- Do not flatten the entire page into one screenshot.
- Text must remain editable, tables must remain editable, and images must remain separate assets.
- The rendered HTML should be visually comparable to this PDF at the same page dimensions.